

Livora Healthcare Platform

Comprehensive Feature Specification

Product & Engineering Documentation

June 7, 2026

Abstract

Livora is an end-to-end digital healthcare platform connecting **patients**, **doctors**, **laboratories**, and **care teams** through a unified web application augmented by an in-house clinical AI assistant (*Livora AI*) and a persistent, cross-session patient memory architecture. This document enumerates every feature available to **doctors** and **patients**, organised by module, and is followed by an audit appendix that confirms each feature's presence in the live codebase and lists additional capabilities discovered during the audit.

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1 Doctor-Facing Features

1.1 Dashboard

- Unified overview combining **practice analytics**, **quick actions**, **today's schedule**, and contextual **highlights** (e.g. profile-completion nudges, pending-request banners surfaced at the top of the page).
- Live stat cards (appointments today, pending requests, completed visits, etc.) and an AI-powered "Smart Schedule Insights" panel.

1.2 Profile & Practice Management

- **Digital signature** upload for use on generated medical documents and prescriptions.
- **Profile picture** and complete personal/professional information, including skills and specialisations.
- **BMDC registration number** entry with verification workflow.
- **Multiple chamber specification** — doctors can register several practice locations, each with **shared map location**.
- **Custom chamber time slots** per chamber — directly drives the time blocks patients see during appointment booking.
- **Turn-by-turn directions** for patients travelling to a chamber (interactive route map).
- **Patient ratings** — aggregate rating and review count visible on the doctor's profile and public listing.

1.3 Appointment Management

- Full lifecycle control: **approve**, **decline/cancel**, and **reschedule** appointment requests.
- **View appointment details** — including the **voice-recorded consultation transcript** and an **AI-generated note/summary** of the visit (generated on demand via a "Quick AI Summary" action, never auto-run).
- **Built-in prescription interface** that launches the moment an appointment/consultation starts.
- Support for both **in-person** and **telemedicine (video consultation)** appointment types.
- Patient-ID badges on every appointment card and the ability to **search appointments directly by patient ID** (in addition to name, email, and phone).

1.4 Patient Directory

- **Auto-generated patient profiles** the moment a doctor approves a new appointment — no manual data entry required.
- **Manual patient profile creation** for patients who cannot self-register; a **temporary-password system** lets the patient log into their own account later and claim the profile.
- **AI-generated condition summary** ("AI Synthesis") shown directly on each patient's card for at-a-glance triage.
- Patient cards display **full personal information** for quick identification.
- **Inside each patient's profile**:
 - Complete history of **previous encounters and appointments**.
 - **Medical documents** uploaded by the patient *and* by the doctor (for patients who cannot upload themselves).
 - **Lab reports** delivered directly from the diagnostic machine/lab.

- **Symptoms History reports** — formal history-taking reports produced by Livora AI from the patient’s own conversations.
- **All medical document types**: discharge reports, SOAP notes, referral letters, medical certificates, consultation notes, and custom reports.
- **MDT rooms and case navigation** — jump straight into any multidisciplinary case or room involving this patient, with a summary view.
- **Critical-condition alert system** — if a patient’s condition is flagged as critical, the doctor can directly **dispatch an emergency alert** to the patient with next-step advice, potentially life-saving.

1.5 Care Team: Multidisciplinary Team (MDT) Support

- **Case creation and resolution** workflow for collaborative, multi-specialist care.
- **Streamlined patient-record navigation** from within a case.
- **MDT rooms** where additional consultants and specialists can be invited, featuring:
 - **Video calling** for real-time multidisciplinary discussion.
 - **One-click full patient history** with **highlighted lab findings and clinical conditions** for faster decision-making.
 - **In-room Livora AI chat** about the patient’s condition — surfaces critical context (medication history, prior clinical conditions) so nothing is missed during the discussion.

1.6 AI Document Generation

- Generate **any kind of medical report or document** about a patient using AI (discharge summaries, referral letters, SOAP notes, medical certificates, consultation notes, operative notes, palliative-care notes, oncology notes, and fully custom reports).
- Generated documents are **automatically routed to that patient’s account** and permanently saved in their Patient Directory profile — nothing is lost.
- **Regenerate** option if the doctor isn’t satisfied with the first draft.

1.7 Independent (Voice) Prescription

- **Voice-prescription support** that records the **entire appointment** audio.
- **Real-time transcription** of the consultation as it happens.
- **Auto-filled prescription** — no manual form filling: medicines, clinical findings, symptoms, diagnosis, recommendations, lab-test orders, and vital statistics are all extracted automatically.
- **Smart intent classification** — the doctor speaks naturally without needing to label or structure their speech; the AI extracts everything required.
- **Multilingual support**: English and Bangla, including **regional dialect support**.
- **Per-medicine clinical suggestions** — while suggesting/confirming medicines, the AI evaluates each one against the patient’s **previous medication history** and flags whether it is a good fit for that specific patient.

1.8 Research Hub

- Advanced **analytics and research support** for doctors, including a research-focused chatbot that is deeply aware of the doctor’s entire patient directory and individual patient data.
- **Literature finding support** — searches medical literature relevant to a query or case.

- **Cohort Builder** — construct patient cohorts from structured criteria for research or audit purposes.
- **Risk Stratification** — AI-assisted classification of patients by clinical risk level.
- **Research Consent** management workflow.
- **Case Report Builder** — assemble structured case reports from patient data.
- **Population Insights** — aggregate, visualised analytics across the doctor's patient population.

1.9 Site-Wide Guided Tour

- A complete, **step-by-step guided walkthrough of every page** on the platform, designed for ease of use.
- Can be **invoked at any time** (not just on first visit) via a persistent tour button.

1.10 Smart Notification System

- Real-time, context-aware notifications covering appointment requests, status changes, lab results, MDT activity, alerts, and more — surfaced consistently across the doctor experience.

2 Patient-Facing Features

2.1 Dashboard

- Personal overview combining **analysis, quick actions, schedule, and highlights**: weekly health check-up prompts, family-history investigation, profile-completion tracking, and recent appointments.
- **Medicine Reminder Matrix** — automatically generated whenever a doctor issues a prescription during an appointment, capturing time, dosage volume, and other details.
- **Zero manual medicine tracking** — an automated reminder system with **one-click medicine-intake logging**.
- **Reminder customisation** — patients can adjust reminder timing and behaviour to fit their routine.

2.2 Personal & Medical Information

- Structured intake of **personal details and medical information** during the patient's first login (onboarding wizard).
- Recurring **weekly health check-ups**.
- Patients can **upload any medical document** they wish directly to their own profile.

2.3 Shareable Health Profile

- **One QR-code scan** lets any doctor, anywhere in the world, instantly view the patient's complete medical history — **nothing missed**: family history, clinical history, health reports, and prescriptions.
- **One-click AI summary** with **critical information highlighted** for faster clinical evaluation by the receiving doctor.
- Eliminates the need to **carry physical documents** and removes the risk of **losing or forgetting** any record — usable anywhere in the world.

2.4 Appointment Booking

- Full **appointment booking** flow with the ability to **view appointment details**.
- Once an appointment is completed, the **prescription is delivered automatically** to the patient's account — it is **never lost**.

2.5 Medical Records

- **AI-based clinical summary** of the patient's records with **highlighted key points** and **plain-English explanations** of health conditions.
- Complete **appointment history**.
- **Medicine tracker with adherence report** — shows how consistently medications are being taken over time.
- **Clinical summary generated from the patient's Livora AI conversations**, with all related documents **downloadable** and shareable via the Shareable Health Profile.

2.6 Multidisciplinary Team (MDT) — Patient View

- **Real-time updates** on case-report progress.
- Visibility into **which specialists** are treating the patient.
- Tracking of **resolved cases**.

2.7 Lab Reports

2.7.1 Overview & Navigation

- Guided tour walking through stats, filters, and the order list.
- Hero header with page title/description and a manual "Lab Matrix Refreshed" data-refresh action.
- Stats dashboard: *Total Tests*, *Pending Payments*, *Completed*, and *Processing* counts at a glance.

2.7.2 Filtering

- Test-type filter: *All* / *Prescribed (doctor-ordered)* / *Self-ordered*.
- Status filter across the full order/test lifecycle (ordered, payment completed, sample collection, completed, reported, etc.).

2.7.3 Self-Service Lab Test Ordering (multi-step modal)

1. **Choose Lab** — browse diagnostic centres/organisations, or (when fulfilling a doctor-prescribed test) see only labs that offer that specific test along with their price.
2. **Select Tests** — search the chosen lab's catalogue and multi-select tests with prices shown.
3. **Confirm & Place Order** — review selected tests and total cost (displayed in BDT, the Bangladeshi Taka currency) before submitting.

2.7.4 Per-Test / Order Cards

- Identity & status: protocol/order number or prescription name, status badge, payment-status badge, sample ID (when assigned).
- Live progress tracker for active orders: *Paid* → *Collecting* → *Analysing* → *Preparing*.
- Diagnostic schema: individual test names, categories, and prices (or clinician/date details for prescribed tests).
- Financial matrix: gross total, amount paid, outstanding balance, with a prompt to clear the balance to speed up results.

2.7.5 Actions per Test

- **Execute Payment** — modal supporting Cash (pay at counter), bKash, or Bank Card; partial-payment entry (minimum 50% required to begin sample processing); transaction-ID capture for online methods; optional notes.
- **Select Lab & Order** — fulfils a doctor-prescribed test by routing into the lab-selection flow.
- **Access Report** — opens completed/reported diagnostic reports via a secure signed URL, supporting multiple files per order.
- **Export Invoice** — generates/downloads an invoice for the order.
- **Cancel Test** — available while an order is still "ordered", with an inline confirm/keep safety step.

2.7.6 Empty / Loading States

- Friendly empty state ("No diagnostic data found") with guidance to adjust filters or place a new order, plus loading skeletons while data is fetched.

2.8 Find Doctors

2.8.1 Search & Discovery

- Text search by doctor name, BMDC registration, or hospital.
- Specialty/department filter via a searchable combobox of medical departments.
- Sorting: *Default*, *Highest Rated*, *Lowest Rated*.

2.8.2 Advanced Filters

- Minimum-rating quick-pick filter.
- Minimum review-count filter.
- Minimum years-of-experience filter.
- One-click "Clear Filters" reset, shown whenever any advanced filter is active.

2.8.3 Location-Based Search ("Near Me")

- "Find doctors near me" — uses browser geolocation to fetch verified doctors sorted by distance within a configurable search radius (km).
- Distance badges ("X km away") and nearest-chamber labels shown on cards while near-me mode is active.
- **Get Directions** — opens an interactive route-map modal from the patient's location to the doctor's chamber.
- "Show All Doctors" toggle to exit near-me mode.

2.8.4 Doctor Cards & Profile Modal

- Cards show photo, name, specialty, rating & review count, availability badge, hospital/practice location, years of experience, BMDC registration number, and consultation fee.
- **View Profile** and **Book Now** actions (booking prompts login first if the user is not authenticated).
- Profile modal: full bio, qualifications, achievements/awards, aggregate & itemised **patient reviews**, and a **Get Directions** action.

2.8.5 Other

- Guided tour covering search/specialty filters, advanced filters, doctor cards, and the booking flow.
- Context-aware empty states (distinct copy for near-me vs. regular search) with one-click reset actions.

2.9 Livora AI — Intelligent Personalised Chatbot

An AI assistant that answers questions based on the patient's **current, real medical status** — not generic advice.

2.9.1 Core Conversational Capabilities

- **Plain-English medical Q&A** via a built-in "Livora Medical Wiki" knowledge search — always reminding the user to consult a real doctor.
- **Personal health-data retrieval**: profile (blood type, allergies, chronic conditions), appointments, prescriptions, active medicines, lab orders/results, and broad medical records.
- **Patient-360 health summary** — aggregates profile, vitals, medications, appointments, recent labs, and vault documents into a holistic "How am I doing?" overview.
- **Document analysis (OCR)** — deep-inspects uploaded PDF/image lab reports and extracts specific values or findings on request.
- **Analytical queries over the patient's own data**, e.g. "How many appointments did I have this year?" or "What antibiotics have I been prescribed?".

2.9.2 Doctor Discovery & Appointment Management (via Chat)

- **Doctor matchmaking** — maps symptoms, specialties, or hospital names to relevant doctors and departments.
- **Find nearest doctors** — uses GPS coordinates (with consent) to rank nearby verified doctors by distance.
- **Booking, cancellation, and rescheduling** of appointments directly through chat, including checking a doctor's live availability/schedule.

2.9.3 Clinical Workflow / Guided History-Taking

- **OSCE-style structured history-taking** — guides the patient through Presenting Complaint, HPI/SOCRATES, Past Medical History, Drug/Family/Social History, Systemic Review, and Ideas-Concerns-Expectations (ICE), then saves a formal PDF report.
- **AI clinical analysis & recommendation** — suggests likely conditions, urgency level, recommended department, and lab tests, then helps the patient pick a specialist and book.

2.9.4 Drug Safety Tools

- **Drug-interaction checker** — looks up official FDA label data for combinations such as "Can I take X with Y?".
- **Dosage information lookup** — official FDA dosing/administration guidance (the AI never invents dosage numbers).

2.9.5 Lifestyle & Wellbeing

- **Conversational lifestyle check-in** — collects smoking, alcohol, exercise, diet, sleep, and stress data and saves it to the patient's lifestyle profile.

2.9.6 Safety & Escalation

- **Emergency detection** — triggers hospital alerts and provides life-saving next steps for acute physical emergencies (chest pain, unconsciousness, severe bleeding, etc.).

- **Mental-health concern flagging** — quietly notifies the care team of self-harm/suicidal disclosures (without an alarming "emergency" framing) and points the patient to crisis-support resources.
- **Identity-mismatch guard** — detects when the person chatting is not the account holder, locks memory for that session, and discloses this to the user.

2.9.7 Memory, Personalisation & Trust Infrastructure

- **Persistent patient memory** — context from prior conversations and mined insights carried across sessions (unless flagged/locked). *See Section 3 for the full architecture.*
- **Conversation summarisation** — recaps prior discussion on request, distinct from a full health-record summary.
- **Workflow/session state tracking** — remembers where the patient is in a multi-step flow (history-taking → analysis → recommendation → doctor selection → booking).
- **Response validation & output constraints** — guards against hallucination, sensitive-data leaks, and prompt injection; the AI never fabricates medical facts and is always tool-backed.
- **Few-shot examples & intent classification** — routes each message to the right tool subset and risk level for efficiency and accuracy.

2.9.8 Channels & Session Management

- **Text chat, streaming responses, and voice input** (voice-to-text runs through the same secure pipeline as text).
- **Conversation history, multiple sessions, feedback submission, history clearing, and session reset.**

3 Persistent Patient Memory — The Signature Innovation

Why It Matters

Stateless LLM calls are the Achilles' heel of every healthcare-AI product: each chatbot turn or "summarise this patient" request starts from zero, re-reading the entire chart and re-deriving insights it already produced — both clinically risky (the AI "forgets" what it told the patient last week) and economically unsustainable (GPT-4o-class costs paid repeatedly for the same answer). Livora replaces this with a **single, bidirectional clinical memory substrate** that both the patient-facing and doctor-facing AI surfaces continuously read from *and* write into — one patient model that gets richer every time either side interacts with the system.

3.1 A Three-Layer Architecture, One Source of Truth

3.1.1 Layer 1 — Structured Memory (the "Facts" Layer)

Each patient carries a versioned JSONB memory object (`aiContext`, schema v2): clinical summary, active medications, key findings, diagnoses, mined chat insights, and medication-adherence patterns — all timestamped and reconciled by an LLM pipeline rather than dumped raw. This cheap, deterministic layer answers *"what do we currently believe about this patient?"*

3.1.2 Layer 2 — Semantic Memory (the "Experience" Layer)

A pgvector-backed vector store (`patient_memory_chunks`, 1536-dimension embeddings, HNSW index) sits on top. Every meaningful update — a new diagnosis, a mined conversational insight, a lab result — becomes embedded chunks. Crucially, Livora uses **hash-diff incremental sync**: only chunks whose content hash actually changed are re-embedded and re-batched, so an unchanged patient costs **zero** embedding tokens on re-sync — the difference between a research prototype and a system that survives a 100,000-patient cost model.

3.1.3 Layer 3 — Intent-Gated Retrieval (not Blanket RAG)

Vector search does *not* fire on every message — that is how teams burn their token budget. Retrieval activates only for intents where memory genuinely changes the answer (personal-data questions, diagnosis discussions, drug/dosage queries, history-taking), using cosine similarity ≥ 0.30 , top-5 results, hard-capped at ~ 520 characters injected into context. Everywhere else, the structured layer alone answers — intent classification is itself a lightweight gate sitting in front of the expensive retrieval path.

3.2 Where It Gets Interesting: The Patient $\leftrightarrow$ Doctor Loop

- **Patient-side signal flows up.** Every chatbot conversation is mined for clinical insights post-session using a small, cheap model — deliberately *not* the expensive frontier-model dispatcher chain. Medication-dosage logs, uploaded lab PDFs, and profile updates all feed a **rate-limited sync orchestrator** that updates shared memory in the background, fire-and-forget, with cooldown windows so the pipeline is never thrashed by noisy input.
- **That same memory then surfaces to the doctor** — not as a raw data dump, but as synthesised intelligence: an `/ai-brief` endpoint returns a pre-computed narrative, diagnosis/medication counts, and adherence-risk flags, read directly from the stored JSONB with

zero additional LLM calls at request time. This replaced what used to be an expensive per-patient bulk fetch with a single lightweight read — because the synthesis work had already happened asynchronously while the patient was using the app.

- **The doctor’s research-hub RAG pipeline blends both worlds.** When a doctor queries the Research Hub, the pipeline retrieves from *both* the document corpus (uploaded records, vault PDFs) *and* `patient_memory_chunks` — citing them side-by-side (`[Doc N]` / `[Mem N]`) within a single context budget. The doctor isn’t getting "what’s in the file" — they’re getting "what’s in the file, *plus* what the patient has actually been telling our AI for months."

3.3 Why It’s Hard to Copy

Most competitors either (a) keep memory client-side / session-only, or (b) bolt on a vector store and let retrieval run unconstrained — simple to build, impossible to run profitably at scale. Livora’s moat is the **engineering discipline wrapped around the substrate**: hash-diff embedding, intent gating, hard token caps on every context block (patient context capped at ~1,200 characters / ~300 tokens; clinical block 750; insights block 370), and cheap-model routing for background extraction versus an LLM fallback chain (GPT-4o → Groq Llama 3.3-70B → NVIDIA → Gemini → OpenRouter) reserved only for what genuinely needs frontier-level reasoning. That discipline is what turns this from a feature into a platform.

Bottom Line

Livora is not selling a chatbot with a database behind it. It is selling a **continuously-compounding clinical memory layer** — the same substrate powers patient self-service, doctor decision support, and population-level synthesis — engineered from day one to be cheap enough to run per-patient, per-day, forever.

4 Audit Appendix — Verification & Refinements

The feature inventory above was cross-checked directly against the live codebase (`client/src/pages`, `server/controllers`, `server/services`). Every feature listed in Sections 1–3 was confirmed to exist as described. This appendix records the audit's refinements and additional capabilities surfaced along the way that were not present in the original feature notes.

4.1 Confirmation Notes

- All ten doctor-side modules (Dashboard, Profile/Practice Management, Appointment Management, Patient Directory, Care Team/MDT, AI Document Generation, Voice Prescription, Research Hub, Site-Wide Tour, Notifications) and all nine patient-side modules were located and verified in the running application.
- **Patient Directory** document types were confirmed against the AI Document Generation engine — the same nine document categories (Discharge, Referral, SOAP, Medical Certificate, Oncology, Specialist Consultation, Operative Note, Palliative Care, Custom) are generated, routed, and archived consistently.
- **Shareable Health Profile**: confirmed QR-token-based public records access (`PublicMedicalRecords`) with AI summarisation and critical-info highlighting; "usable globally without physical documents" is an accurate characterisation of the underlying public-link mechanism.
- Two small, recent refinements verified live on the platform: the doctor dashboard's "Pending Appointment Requests" banner was relocated from the bottom of the page to immediately below the welcome header (so it's seen first), and the doctor appointments view now displays each patient's **Patient ID** on their card and supports **searching appointments by Patient ID** directly.

4.2 Additional Features Discovered During the Audit

- **Doctor Ratings & Reviews subsystem** — a complete patient-feedback and rating pipeline (including an admin moderation view) feeding the rating displays seen on doctor cards, profiles, and the dashboard.
- **Integrated payments & billing** — a multi-method payment system (Cash / bKash / Bank Card) spanning both consultation fees and lab-test payments, with partial-payment support, transaction-ID capture, and invoice export.
- **Lab-report OCR extraction pipeline** — automatic structured-data extraction from uploaded lab PDFs/images, powering both the chatbot's "analyze a report" capability and the Patient Directory's lab views.
- **Vital-signs capture** (blood pressure, heart rate, temperature, etc.) embedded directly into the prescription workflow and surfaced through medical records and AI summaries.
- **Medicine adherence analytics** — a dedicated computation pipeline (running on the same persistent-memory substrate) that produces per-medicine adherence rates and "running low" / "low adherence" flags, surfaced to both patients (Medicine Tracker) and doctors (AI Synthesis briefs).
- **AI-quick-summary actions are opt-in by design** — across the platform (e.g. consultation transcripts, document overviews) the AI generates summaries only when explicitly requested via a button, never automatically, preserving doctor control and minimising unnecessary inference cost.

4.3 Audit Verdict

Area	Status
Doctor-side feature list (10 modules)	Verified — accurate
Patient-side feature list (9 modules)	Verified — accurate
Persistent Patient Memory architecture	Verified — accurate
Additional capabilities found	6 items — documented above

No claimed feature was found to be missing, fabricated, or materially inaccurate. The refinements above are additive — they extend, not contradict, the original feature inventory.